

Lecture Notes: Applications of Derivatives

Calculus I

Author: Bakdaulet Sotsial

Astana IT University

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Section 1 Maxima and Minima

Definition 1 (Absolute Maximum and Minimum) Let f be a function defined on a set D containing c . Then $f(c)$ is the absolute maximum of f on D if $f(c) \geq f(x)$ for all $x \in D$. Similarly, $f(c)$ is the absolute minimum of f on D if $f(c) \leq f(x)$ for all $x \in D$. The absolute maximum and minimum are called the absolute extrema of f on D .

Definition 2 (Local Maximum and Minimum) A function f has a local maximum at c if $f(c) \geq f(x)$ for all x in an open interval containing c . Similarly, f has a local minimum at c if $f(c) \leq f(x)$ for all x in an open interval containing c . Local maxima and minima are called local extrema.

Theorem 1 (Extreme Value Theorem) If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum and an absolute minimum on $[a, b]$.

Theorem 2 (Fermat's Theorem) If f has a local extremum at c and $f'(c)$ exists, then $f'(c) = 0$.

Suppose f has a local maximum at c . Then for h sufficiently small (positive or negative), $f(c+h) \leq f(c)$, so $f(c+h) - f(c) \leq 0$. For $h > 0$, $\frac{f(c+h)-f(c)}{h} \leq 0$, so the right-hand derivative is ≤ 0 . For $h < 0$, $\frac{f(c+h)-f(c)}{h} \geq 0$, so the left-hand derivative is ≥ 0 . Since $f'(c)$ exists, both one-sided limits must be equal, so $f'(c) = 0$. The proof for a local minimum is similar.

Definition 3 (Critical Number) A critical number of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

Remark 1 If f has a local extremum at c , then c is a critical number of f . However, not every critical number yields a local extremum. For example, $f(x) = x^3$ has $f'(0) = 0$, but $x = 0$ is neither a local maximum nor a local minimum.

Theorem 3 (First Derivative Test) Suppose c is a critical number of a continuous function f .

1. If f' changes from positive to negative at c , then f has a local maximum at c .
2. If f' changes from negative to positive at c , then f has a local minimum at c .
3. If f' does not change sign at c , then f has no local extremum at c .

Theorem 4 (Second Derivative Test) Suppose f'' is continuous near c .

1. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
2. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .
3. If $f'(c) = 0$ and $f''(c) = 0$, the test is inconclusive.

Example 1 Find the local extrema of $f(x) = x^3 - 3x^2 + 2$. First, find critical numbers: $f'(x) = 3x^2 - 6x = 3x(x-2)$. Setting $f'(x) = 0$ gives $x = 0$ and $x = 2$. Second derivative: $f''(x) = 6x - 6$. At $x = 0$: $f''(0) = -6 < 0$, so local maximum at $x = 0$, $f(0) = 2$. At $x = 2$: $f''(2) = 6 > 0$, so local minimum at $x = 2$, $f(2) = 8 - 12 + 2 = -2$.

Theorem 5 (Closed Interval Method) To find the absolute extrema of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b) .

2. Find the values of f at the endpoints a and b .
3. The largest of these values is the absolute maximum; the smallest is the absolute minimum.

Example 2 Find the absolute extrema of $f(x) = x^3 - 3x + 1$ on $[0, 3]$. $f'(x) = 3x^2 - 3 = 3(x^2 - 1)$. Critical numbers in $(0, 3)$: $x = 1$. Evaluate: $f(0) = 1$, $f(1) = -1$, $f(3) = 27 - 9 + 1 = 19$. Absolute maximum: 19 at $x = 3$. Absolute minimum: -1 at $x = 1$.

Section 2 The Mean Value Theorem

Theorem 6 (Rolle's Theorem) If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists a number $c \in (a, b)$ such that $f'(c) = 0$.

If f is constant on $[a, b]$, then $f'(c) = 0$ for all $c \in (a, b)$. Otherwise, by the Extreme Value Theorem, f attains a maximum or minimum at some $c \in (a, b)$. Since $f(a) = f(b)$, this extremum is not at the endpoints, so $c \in (a, b)$. By Fermat's Theorem, $f'(c) = 0$.

Theorem 7 (Mean Value Theorem) If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Equivalently, $f(b) - f(a) = f'(c)(b - a)$.

Define $g(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$. Then g is continuous on $[a, b]$, differentiable on (a, b) , and $g(a) = 0 = g(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ such that $g'(c) = 0$. Since $g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$, we get $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Example 3 Verify the Mean Value Theorem for $f(x) = x^2$ on $[0, 2]$. $f(0) = 0$, $f(2) = 4$. The average rate of change is $\frac{4 - 0}{2 - 0} = 2$. We need $f'(c) = 2c = 2$, so $c = 1 \in (0, 2)$.

Theorem 8 (Consequences of the Mean Value Theorem) 1. If $f'(x) = 0$ for all x in an interval (a, b) , then f is constant on (a, b) .

2. If $f'(x) = g'(x)$ for all x in (a, b) , then $f(x) = g(x) + C$ for some constant C .

3. If $f'(x) > 0$ for all x in an interval, then f is increasing on that interval.

4. If $f'(x) < 0$ for all x in an interval, then f is decreasing on that interval.

For (1): Let $x_1 < x_2$ be any two points in (a, b) . By the Mean Value Theorem, there exists $c \in (x_1, x_2)$ such that $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$. Thus $f(x_2) = f(x_1)$, so f is constant. The others are similar.

Section 3 Graphing by the First and Second Derivatives

Definition 4 (Concavity) A function f is concave up on an interval I if f' is increasing on I . It is concave down on I if f' is decreasing on I .

Theorem 9 (Test for Concavity) 1. If $f''(x) > 0$ for all x in I , then f is concave up on I .

2. If $f''(x) < 0$ for all x in I , then f is concave down on I .

Definition 5 (Inflection Point) A point $(c, f(c))$ on a curve is an inflection point if the curve changes from concave up to concave down, or vice versa, at that point.

Theorem 10 (Second Derivative Test for Inflection Points) If $f''(c) = 0$ and f'' changes sign at c , then $(c, f(c))$ is an inflection point.

Example 4 Analyze $f(x) = x^4 - 4x^3$. $f'(x) = 4x^3 - 12x^2 = 4x^2(x - 3)$. Critical numbers: $x = 0, 3$. $f''(x) = 12x^2 - 24x = 12x(x - 2)$. Possible inflection points: $x = 0, 2$. Intervals for f' : $(-\infty, 0)$: $f' < 0$ (decreasing); $(0, 3)$: $f' < 0$ (decreasing); $(3, \infty)$: $f' > 0$ (increasing). Local minimum at $x = 3$, $f(3) = 81 - 108 = -27$. Intervals for f'' : $(-\infty, 0)$: $f'' > 0$ (concave up); $(0, 2)$: $f'' < 0$ (concave down); $(2, \infty)$: $f'' > 0$ (concave up). Inflection points at $x = 0$, $f(0) = 0$ and $x = 2$, $f(2) = 16 - 32 = -16$.

Theorem 11 (Summary of Graphing with Derivatives) To graph f :

1. Find the domain of f .
2. Find the x - and y -intercepts.
3. Find vertical and horizontal asymptotes.
4. Find f' and critical numbers; determine intervals of increase/decrease.
5. Find f'' and possible inflection points; determine concavity.
6. Plot key points and sketch the curve.

Section 4 Asymptotes and Dominant Terms

Definition 6 (Vertical Asymptote) The line $x = a$ is a vertical asymptote of $y = f(x)$ if at least one of the following holds: $\lim_{x \rightarrow a^-} f(x) = \pm\infty$, $\lim_{x \rightarrow a^+} f(x) = \pm\infty$, or $\lim_{x \rightarrow a} f(x) = \pm\infty$.

Definition 7 (Horizontal Asymptote) The line $y = L$ is a horizontal asymptote of $y = f(x)$ if either $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.

Definition 8 (Slant Asymptote) The line $y = mx + b$ (with $m \neq 0$) is a slant asymptote of $y = f(x)$ if

$$\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0 \quad \text{or} \quad \lim_{x \rightarrow -\infty} [f(x) - (mx + b)] = 0.$$

Theorem 12 (Finding Slant Asymptotes) A rational function $f(x) = \frac{P(x)}{Q(x)}$ has a slant asymptote if the degree of P is exactly one more than the degree of Q . The slant asymptote is the quotient obtained by polynomial long division.

Definition 9 (Dominant Terms) For a rational function $f(x) = \frac{P(x)}{Q(x)}$ as $x \rightarrow \pm\infty$, the dominant term of the numerator is its highest-degree term, and the dominant term of the denominator is its highest-degree term. The end behavior of f is determined by the ratio of these dominant terms.

Example 5 Find the asymptotes of $f(x) = \frac{2x^2+3x-1}{x^2-4}$. Vertical asymptotes: set denominator $x^2 - 4 = 0$, so $x = \pm 2$. Horizontal asymptote: degrees are equal (both 2). The leading coefficients are 2 and 1, so $y = 2$ is a horizontal asymptote. No slant asymptote since degrees are equal.

Example 6 Find the asymptotes of $f(x) = \frac{x^2+1}{x-1}$. Vertical asymptote: $x = 1$. Slant asymptote: degree of numerator is 2, degree of denominator is 1, so a slant asymptote exists. Perform polynomial division:

$$\frac{x^2+1}{x-1} = x+1 + \frac{2}{x-1}.$$

As $x \rightarrow \pm\infty$, $\frac{2}{x-1} \rightarrow 0$, so the slant asymptote is $y = x+1$.

Section 5 Graphing Rational Functions

Definition 10 (Rational Function) A rational function is a function of the form $f(x) = \frac{P(x)}{Q(x)}$, where P and Q are polynomials and $Q(x) \neq 0$.

Theorem 13 (Steps for Graphing a Rational Function) 1. Factor the numerator and denominator.

2. Find the domain and vertical asymptotes (where denominator is zero but numerator is not).
3. Find holes (where both numerator and denominator are zero).
4. Find horizontal or slant asymptotes using dominant terms.
5. Find x - and y -intercepts.
6. Analyze the first derivative for intervals of increase/decrease and local extrema.
7. Analyze the second derivative for concavity and inflection points.
8. Sketch the graph.

Example 7 Graph $f(x) = \frac{x^2-1}{x^2-4}$. Factor: $f(x) = \frac{(x-1)(x+1)}{(x-2)(x+2)}$. Domain: $x \neq \pm 2$. Vertical asymptotes: $x = 2$ and $x = -2$. No holes. Horizontal asymptote: degrees equal, leading coefficients 1 and 1, so $y = 1$. Intercepts: x -intercepts at $x = \pm 1$; y -intercept at $f(0) = \frac{-1}{-4} = \frac{1}{4}$. First derivative:

$$f'(x) = \frac{2x(x^2-4) - (x^2-1)(2x)}{(x^2-4)^2} = \frac{2x[(x^2-4) - (x^2-1)]}{(x^2-4)^2} = \frac{2x(-3)}{(x^2-4)^2} = \frac{-6x}{(x^2-4)^2}.$$

Critical number: $x = 0$. $f' > 0$ for $x < 0$, $f' < 0$ for $x > 0$, so local maximum at $x = 0$, $f(0) = \frac{1}{4}$. Second derivative can be computed, but the graph is now well understood.

Section 6 Related Rates of Change

Definition 11 (Related Rates) Related rates problems involve finding the rate of change of one quantity in terms of the rate of change of another quantity. The key is to find an equation relating the quantities and then differentiate both sides with respect to time t using the chain rule.

Theorem 14 (Strategy for Related Rates Problems) 1. Draw a diagram and label quantities.

2. Write an equation relating the variables.
3. Differentiate both sides with respect to t .

4. Substitute known values and solve for the unknown rate.

Example 8 A spherical balloon is being inflated at a rate of $100 \text{ cm}^3/\text{s}$. How fast is the radius increasing when the radius is 5 cm ? Volume of sphere: $V = \frac{4}{3}\pi r^3$. Differentiate with respect to t :

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$

Given $\frac{dV}{dt} = 100$ and $r = 5$:

$$100 = 4\pi(25) \frac{dr}{dt} \implies \frac{dr}{dt} = \frac{100}{100\pi} = \frac{1}{\pi} \text{ cm/s}.$$

Example 9 A ladder 10 ft long rests against a vertical wall. If the bottom of the ladder slides away from the wall at 1 ft/s , how fast is the top sliding down the wall when the bottom is 6 ft from the wall? Let x be the distance from the wall to the bottom, y be the height of the top. By Pythagoras: $x^2 + y^2 = 100$. Differentiate with respect to t :

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0 \implies \frac{dy}{dt} = -\frac{x}{y} \frac{dx}{dt}.$$

Given $\frac{dx}{dt} = 1$ and $x = 6$. Then $y = \sqrt{100 - 36} = 8$. So

$$\frac{dy}{dt} = -\frac{6}{8} \cdot 1 = -\frac{3}{4} \text{ ft/s}.$$

The top is sliding down at 0.75 ft/s .

Section 7 Optimization

Definition 12 (Optimization Problem) An optimization problem is a problem in which we need to find the maximum or minimum value of a function subject to certain constraints.

Theorem 15 (Strategy for Optimization) 1. Understand the problem and identify the quantity to be optimized.

2. Draw a diagram if appropriate.

3. Introduce variables and write an equation for the quantity.

4. Express the quantity in terms of a single variable using constraints.

5. Find the domain of the variable.

6. Find critical numbers and use the first or second derivative test to find extrema.

7. Verify the solution answers the question.

Example 10 Find two numbers whose sum is 20 and whose product is as large as possible. Let the numbers be x and y . Constraint: $x + y = 20 \implies y = 20 - x$. Product: $P = xy = x(20 - x) = 20x - x^2$. $P'(x) = 20 - 2x = 0 \implies x = 10$. $P''(x) = -2 < 0$, so maximum. The numbers are 10 and 10 , product 100 .

Example 11 A rectangular box with a square base and no top must have a volume of 32 cm^3 . Find the dimensions that minimize the amount of material used. Let x be the side of the base and h be the height. Volume: $x^2h = 32 \implies h = \frac{32}{x^2}$. Surface area (no top): $S = x^2 + 4xh = x^2 + 4x \cdot \frac{32}{x^2} = x^2 + \frac{128}{x}$. $S'(x) = 2x - \frac{128}{x^2} = 0 \implies 2x^3 = 128 \implies x^3 = 64 \implies x = 4$. $S''(x) = 2 + \frac{256}{x^3} > 0$ for $x > 0$, so minimum. Then $h = \frac{32}{16} = 2$. Dimensions: $4 \times 4 \times 2 \text{ cm}$.

Section 8 Indeterminate Forms and L'Hôpital's Rule

Definition 13 (Indeterminate Forms) A limit is called an indeterminate form if direct substitution yields an expression such as $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \cdot \infty$, $\infty - \infty$, 0^0 , 1^∞ , or ∞^0 .

Theorem 16 (L'Hôpital's Rule) Suppose f and g are differentiable on an open interval containing a (except possibly at a), and $g'(x) \neq 0$ near a . If

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0,$$

or if

$$\lim_{x \rightarrow a} f(x) = \pm\infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm\infty,$$

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided the limit on the right exists or is $\pm\infty$. The rule also applies for one-sided limits and as $x \rightarrow \pm\infty$.

Example 12 Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{x}$. Direct substitution gives $\frac{0}{0}$. Apply L'Hôpital's Rule:

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1.$$

Example 13 Evaluate $\lim_{x \rightarrow \infty} \frac{\ln x}{x}$. This is of the form $\frac{\infty}{\infty}$. Apply L'Hôpital's Rule:

$$\lim_{x \rightarrow \infty} \frac{\ln x}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0.$$

Example 14 Evaluate $\lim_{x \rightarrow 0^+} x \ln x$. This is of the form $0 \cdot (-\infty)$. Rewrite as $\frac{\ln x}{1/x}$, which is $\frac{-\infty}{\infty}$. Apply L'Hôpital's Rule:

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{1/x} = \lim_{x \rightarrow 0^+} \frac{1/x}{-1/x^2} = \lim_{x \rightarrow 0^+} (-x) = 0.$$

Example 15 Evaluate $\lim_{x \rightarrow 0^+} x^x$. This is of the form 0^0 . Let $y = x^x$, so $\ln y = x \ln x$. Then

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln x = 0 \quad (\text{from previous example}).$$

Thus, $\lim_{x \rightarrow 0^+} x^x = e^0 = 1$.

Theorem 17 (Other Indeterminate Forms) For forms $\infty - \infty$, 0^0 , 1^∞ , and ∞^0 , first transform the expression into a quotient or use logarithms to convert to a form where L'Hôpital's Rule can be applied.